

Ruting Wang

Finance PhD specializing in risk modelling, digital assets, sustainability economics and causal ML

rutiwang@cityu.edu.hk • wangrt6@mail.sysu.edu.cn • rutingrainywang.com • Google Scholar • ORCID

PROFESSIONAL SUMMARY

Finance researcher with a PhD in Finance and postdoctoral experience at City University of Hong Kong and Sun Yat-sen University. I build empirical tools for measuring financial risk, digital-asset market dynamics, carbon exposure and institutional shocks, using machine learning, causal inference, network methods and time-series modelling. My work is suited to quantitative research, risk analytics, ESG and climate-risk analytics, digital-assets research, market intelligence and policy-facing data science roles.

CORE CAPABILITIES

Risk analytics	Systemic risk, tail dependence, Financial Risk Meter, bond-market risk, institutional shock transmission.
Machine learning	Quantile regression, Lasso and regularisation, double machine learning, forecasting, model validation.
Causal inference	Policy evaluation, quasi-experimental design, institutional reform, capital allocation and firm behaviour.
Digital assets	Cryptocurrency risk, Ethereum protocol events, pricing-kernel volatility, market response and energy use.
Sustainability	Carbon-risk dynamics, climate-policy asymmetry, stranded-asset risk and resource reallocation.

SELECTED APPLIED PROJECTS

Financial Risk Meter for China	Systemic risk / network analytics
Designed a China-focused systemic-risk indicator using tail-event dependence and quantile-Lasso methods to identify risk transmission among major financial institutions and macro-financial drivers.	
FRM for Crypto Markets	Digital assets / market risk
Extended network-risk measurement to cryptocurrency markets, linking Financial Risk Meter signals with pricing-kernel volatility and forward-looking market risk.	
Carbon Risk Dynamic Indicator (CARDI)	Climate finance / ESG analytics
Built a Quantile-LASSO network indicator to measure carbon-risk transmission, volatility-demand shifts and low-carbon premia across financial markets.	
Ethereum Development Milestones	Digital assets / event analytics
Evaluated market responses to Ethereum protocol upgrades, including return, volatility and spillover effects around major development milestones.	
Bond-Market Default Risk and Resource Allocation	Credit risk / causal inference
Studied how rigid payment expectations, implicit guarantees and default-risk networks affect bond-market pricing efficiency and financial-resource allocation.	

EXPERIENCE

2024.12-present	Postdoctoral Researcher , Laboratory of Energy Economics and Environmental Management, School of Energy and Environment, City University of Hong Kong. Hong Kong Scholars Program.
2023.02-present	Postdoctoral Researcher , Business School, Sun Yat-sen University. Research on bond-market risk, institutional reform, machine-learning causal inference and capital allocation.
2023.12-present	Research Fellow , IDA Institute for Digital Assets, Bucharest University of Economic Studies. Research on digital assets, financial risk and market design.
2021.01-2022.01	Visiting PhD Researcher , Ladislaus von Bortkiewicz Chair of Statistics, Humboldt-Universität zu Berlin. Research on risk measurement and quantitative methods.

SELECTED PUBLICATIONS AND OUTPUTS

- Wang, R., Poti, V. and Haerdle, W. K. (2024). Assessing network risk with FRM: Links with pricing kernel volatility and application to cryptocurrencies. *Quantitative Finance*.

- **Wang, R.**, Althof, M. and Haerdle, W. K. (2023). A financial risk meter for China. *Emerging Markets Review*.
- Chen, R., Liu, S., Nie, Z. and **Wang, R.** (2026). Bank deregulation and the rise of local government debt. *Economic Modelling*.
- **Wang, R.**, Haerdle, W. K. and Zhang, L. Measuring carbon risk dynamics. Working paper under review.
- Huang, C., Bag, R., Pele, P., Petukhina, A., **Wang, R.** and Zuo, X. *Applied Time Series Analysis and Forecasting with Python: Problems and Solutions*. Springer Nature, forthcoming.

TECHNICAL TOOLKIT

Programming	Python, R, Stata, MATLAB, TEX .
Methods	Quantile regression, Lasso, double machine learning, network analysis, Kalman filtering, bootstrap and jackknife, event-study design, panel-data analysis, time-series forecasting.
Domains	Financial institutions, credit and bond markets, crypto assets, ESG and carbon risk, institutional reform, corporate finance, resource allocation.

TEACHING, COMMUNICATION AND SERVICE

Open learning	Created Quantinar courses and courselets on machine learning in financial risk, FRM@China, FRM@Crypto, carbon-risk dynamics, Ethereum milestones, Kalman filtering, quantile regression and time-series methods.
Teaching	Machine Learning in Financial Risk, Romania summer course; Digital Economy and Decision Analytics, Humboldt-Universität zu Berlin; Probability Theory and Financial Markets, Sun Yat-sen University.
Editorial	Associate Editor, <i>Digital Finance</i> ; Associate Editor and Special Issue Editor, <i>Management & Marketing</i> , Springer Nature special issue on Generative AI in Finance.

EDUCATION

2018.09-2022.12	PhD in Finance , Sun Yat-sen University. Dissertation on rigid payment, debt financing, capital misallocation and machine-learning causal inference.
2014.09-2017.06	MSc in Financial Management , Zhongnan University of Economics and Law.
2010.09-2014.06	BSc in Financial Management , Zhongnan University of Economics and Law.